AST 6215 Stars and the Galaxy 2026
Problem Set #2
Due Sep 15, 2026 4am

Useful numbers:
Solar M 1.989 x 10^{33} g Solar L 3.844 x 10^{33} erg/s Solar R 6.96 x 10^{10} cm

As always, please list your collaborators and resources used.

1. Superadiabaticity:

$$F_{conv} = \rho c_p T \left(\frac{l_m}{H_p}\right)^2 \sqrt{\left(\frac{1}{2} g H_p\right)} (\nabla - \nabla_{ad})^{3/2} = \frac{L}{4\pi r^2}$$

Assume that all the flux in the sun was carried by convection. You can simplify the calculation by taking average values for everything.

$$\bar{\rho} = \frac{3M}{4\pi R^3} \qquad \bar{T} = \frac{\mu}{R_{gas}} \frac{GM}{R} \qquad \bar{P} = \frac{GM^2}{R^4} \qquad c_p = \frac{5}{2} \frac{R_{gas}}{\mu} \qquad H_p = \frac{P}{\rho g}$$

a) In this simple approach, what should you assume for l_m ?
b) Express F_{conv} in terms of M, R, G and $\nabla - \nabla_{ad}$. You may drop any constants of order unity.
c) Assume all of the flux in the star is carried by convection, solve for $\nabla - \nabla_{ad}$ in terms of M, R, G and L.
d) How much steeper than adiabatic does the true gradient need to be to transport all the flux? Comment on what that means physically.

2. Calculate the ratio of H^- to neutral hydrogen for T=6000K and an electron pressure of 30 dynes/cm^2. Assume all the atoms in a given ionization state are in the ground state of that ion. The ionization potential of H^- is 0.7 eV.

3. The first ionization potential of hydrogen is 13. 6 eV. The subsequent energies scale with $1/n^2$, with each succeeding energy level being easier to ionize.
a) Compute the ionization energies of the first 7 levels of hydrogen.
b) For a wavelength near 6000 Angstroms, what levels of hydrogen can contribute to the bound-free opacity for a material at 6000 K? What sets the minimum and maximum values?

4). Nuclear Fusion
a) Make a quantative and qualitative argument that the fraction of H atoms moving fast enough to fuse in the **classical** limit is approximately 10^{-434}. (I'm fine with you not actually doing the ugly integral if you can explain why without doing that)
b) Using this probability, make an argument for how much hydrogen would be fused in the Sun over the solar lifetime.
c) Explain in words what is missing from the classical approximation, and how it allows the sun to shine but not burn out immediately.

5) YREC

a) Use a solar mass model run to about the solar age and parameters. Using the Convective flag ‘C’ in the .last file, identify the location in the interior where the star switches from radiative to convective. Why does this occur?
b) Use the temperature and pressure in these shells to compute the true temperature gradient ∇ . How does it compare to your expectations of the adiabatic gradient ∇_{ad} ? Is this consistent or inconsistent with what we discussed in class and why?